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Inference of multicast routing trees and bottleneck bandwidths using end-to-end measurements

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Abstract

The efficacy of end-to-end multicast transport protocols depends critically upon their ability to scale efficiently to a large number of receivers. Several research multicast protocols attempt to achieve this high scalability by identifying sets of co-located receivers in order to enhance loss recovery, congestion control and so forth. A number of these schemes could be enhanced and simplified by some level of explicit knowledge of the topology of the multicast distribution tree, the value of the bottleneck bandwidth along the path between the source and each individual receiver and the approximate location of the bottlenecks in the tree. In this paper, we explore the problem of inferring the internal structure of a multicast distribution tree using only observations made at the end hosts. By noting correlations of loss patterns across the receiver set and by measuring how the network perturbs the fine-grained timing structure of the packets sent from the source, we can determine both the underlying multicast tree structure as well as the bottleneck bandwidths. The simulations show that the algorithm is robust and appears to converge to the correct tree with high probability.

Real-World Impact & Application

Early Internet Video Streaming & Telemetry: During the creation of the Multicast Backbone (MBone) for live video and audio broadcasts, network engineers had no way to pinpoint which router links were congested. Ratnasamy & McCanne invented end-to-end probing to estimate bottleneck bandwidths and routing trees, laying the technical foundation for modern adaptive video streaming (HLS, DASH).